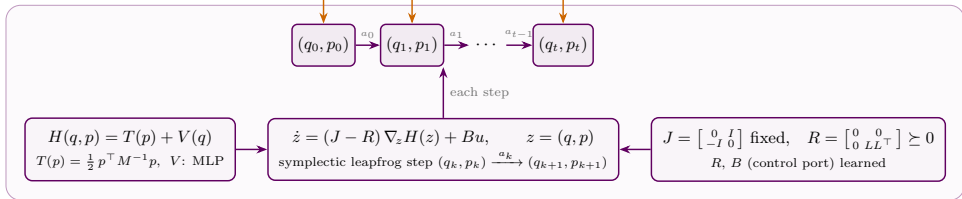
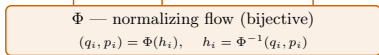
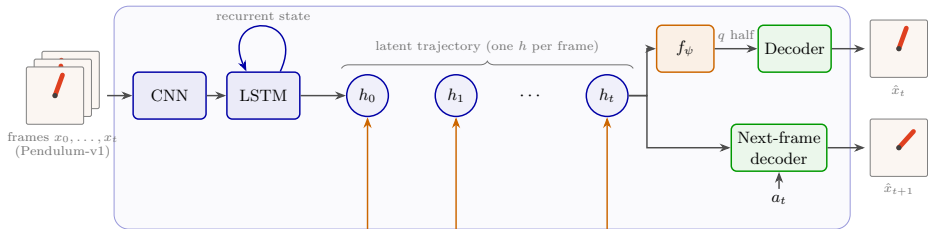


Phase 1 — LSTM autoencoder (pixel space)

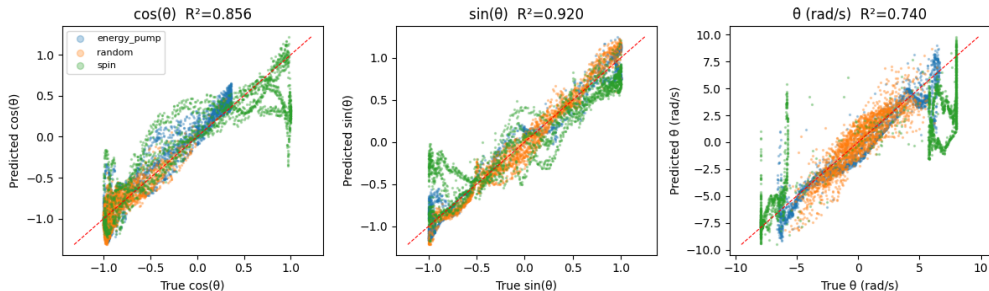


Phase 2 — port-Hamiltonian dynamics (learned phase space)

Does $h_0, \dots, h_t$ encode angle and angular velocity?

Linear probe: least-squares fit $h_t \mapsto (\cos \theta, \sin \theta, \dot{\theta})$, evaluated on held-out trajectories.

Latent $\rightarrow$ state regression, held-out trajectories (epoch 5000)



$\Rightarrow$ the pendulum's phase space $(\theta, \dot{\theta})$ is linearly decodable from h .